

Question	Answer	Marks
9(a)	(magnetic) flux density $\times$ area	B1
	magnetic flux density normal to area or reference to cross-sectional area or $\times \sin$ (angle between B and A)	B1
	$\times$ number of turns on coil	B1
9(b)	e.m.f. = BAN / t or e.m.f. = rate of change of flux <u>linkage</u>	C1
	$= (7.5 \times 10^{-3} \times \pi \times \{1.2 \times 10^{-2}\}^2 \times 160) / 0.15$ $= 3.6 \times 10^{-3} \text{ V}$	A1
9(c)	sketch: zero for 0–0.10 s, 0.25–0.35 s, and 0.425–0.55 s, and non-zero outside these ranges	B1
	two horizontal steps, with zero voltage either side	B1
	with same polarity	B1
	correct values (1st step 3.6 mV and 2nd step 7.2 mV)	B1

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10(c)	emission of electron	B1